FGV EMAp
ESCOLA DE MATEMÁTICA APLICADA

Machine Learning

LAB EXERCISE 4

This lab exercise must be submitted by May 23rd, 2025 at 10:00 am.
Late submissions will not be accepted and will be marked as zero.

This lab exercise is mandatory and could be assessed.
It is worth 1% of your total final grade for this course.

Submission Instructions

Submit your files through EClass. The files you submit cannot be read by any other students. You can replace your submission as many times as you like by resubmitting it, although only the last version sent is kept.

If you have last-minute problems with your EClass submission, email your assignment as an attachment to alberto.paccanaro@fgv.br with the subject "URGENT – LAB 4 SUBMISSION". In the body of the message, explain the reason for not submitting it through EClass.

All the work you submit should be solely your own work.
Lab exercise submissions will be checked for this.

In this lab, you will implement a neural network model for a classification task using numpy. You will implement the forward pass and backpropagation algorithms for training your model. After your model is trained, you will evaluate its performance using the Receiver Operating Characteristic (ROC) curve. The datasets for this lab are contained in the files "Raisin_train.csv" and "Raisin_test.csv".

DATASET DESCRIPTION

1. ***Raisin dataset*:** different varieties of raisins are grown in Turkey and that belong to 2 classes, Kecimen and Besni raisin. For this dataset, a total of 900 raisin varieties were used. Original images were subjected to various stages of pre-processing and 7 morphological features were extracted.

   Attribute Information:

   - **Area:** Gives the number of pixels within the boundaries of the raisin.
   - **Perimeter:** It measures the environment by calculating the distance between the boundaries of the raisin and the pixels around it.
   - **MajorAxisLength:** Gives the length of the main axis, which is the longest line that can be drawn on the raisin.
   - **MinorAxisLength:** Gives the length of the small axis, which is the shortest line that can be drawn on the raisin.
   - **Eccentricity:** It gives a measure of the eccentricity of the ellipse, which has the same moments as raisins.
   - **ConvexArea:** Gives the number of pixels of the smallest convex shell of the region formed by the raisin.
   - **Extent:** Gives the ratio of the region formed by the raisin to the total pixels in the bounding box.
   - **Class:** Kecimen and Besni raisin.

   The datasets contains 900 datapoints, the first 7 columns are the features, and the one to be predicted is the last one.

YOUR TASKS

A) Implement a function to train a neural network using the training set: this function should learn the weights of the neural network by stochastic gradient descent, calculating gradients using backpropagation. Your neural network will output the probability that a raising belongs to the Besni species. This function should be flexible and be able to accept any number of layers and neurons per layer as an input parameter. Make sure to include weight decay in your loss function.

B) Make predictions with your trained neural network: implement a function that predicts the probability that a raising is from the Besni species using your trained neural network. You should evaluate your results by visualizing the Receiver Operating Characteristic (ROC) curve and computing the Area under that curve.

TIP 1: it will be helpful to plot the error function vs the iteration.

TIP 2: to initialize the weights, you can choose them randomly, gaussian distributed, with a very small variance.

Run some experiments and think about these questions:

- Do this implementation always converge to the same solution?
- What happens if the step size of your stochastic gradient descent is too large?
- What happens if you initialize your weights to values that are too large?
- What happens if your neural network is too deep?
- How can you select a threshold for your neural network output?

Have fun ! 😊